Yongda Zhu, Ph.D.
Steward Observatory, University of Arizona,
933 N Cherry Ave, Tucson, AZ 85721, USA
yongdaz@arizona.edu | ydzhuastro.github.io

RESEARCH PROFILE

I study the **phase transition** of the Universe during cosmic **reionization** and how **galaxies** and **quasars** exchange mass, metals, and ionizing radiation with the **intergalactic medium**, which contains most of the Universe's baryonic matter. I combine quasar absorption spectroscopy, **JWST**, **ALMA**, **Keck**, **Roman**, and cosmological simulations to connect galactic-scale physics with the evolution of the **cosmic baryon cycle**.

My work has constrained the **timing and topology of late reionization**, established links between IGM transmission and galaxy environment, uncovered signatures of quasar feedback on intergalactic scales, revealed **metal-enriched baryon cycling** only 500 Myr after the Big Bang, and developed **AI-enabled imaging approaches** to studying the baryon cycle with Roman and JWST.

Selected record: Ph.D. in Physics (**2024**); **16** first-author papers, including publications in *ApJ*, *ApJL*, *Nature Astronomy*, and *MNRAS*; **3,500+** total citations (**430+** first-author citations); **3** student-led papers; PI/co-PI on **4** awarded Roman programs totaling **$1.2M** (**$0.6M** in direct PI support and **$0.6M** supporting collaborators through co-PI programs); PI of ALMA programs totaling **>40 hr** and more than **30 nights** on ground-based telescopes, including Keck and MMT; data or analysis leadership across approximately **2,000 hr** of competitively allocated JWST observations.

ACADEMIC APPOINTMENTS

- **Assistant Research Professor**, University of Arizona Oct. 2026 – present
  Leading research programs investigating cosmic baryon cycles with Roman and JWST.
- **JASPER Postdoctoral Scholar**, University of Arizona Aug. 2025 – Oct. 2026
  Mentors: Prof. Eiichi Egami & Prof. Xiaohui Fan
  Leading NIRCam/WFSS analysis for SAPPHIRES and studying gas–galaxy interactions across scales with JWST and machine learning.
- **Postdoctoral Research Associate**, NIRCam contract Feb. 2024 – Aug. 2025
  Mentors: Prof. Marcia Rieke & Prof. George Rieke
  Member of JWST NIRCam Science Team and MIRI U.S. Team.
  Led a systematic search for galaxies with extended line emission and candidate outflows using deep NIRCam medium-band imaging and machine learning.
  Leading the SMILES (JWST-GTO-1207) NIRSpec public data release and multi-band NIRCam photometric analysis of galaxies.
  Leading reduction and analysis of JWST/NIRSpec MSA spectroscopy of cosmic-noon galaxies and obscured AGN.
- **Graduate Student Researcher**, UC Riverside Jul. 2019 – Feb. 2024
  Mentor: Prof. George Becker

Led Keck and ALMA observations of the IGM via high-z quasars; modeled the Lyα and Lyβ forest using cosmological simulations.
Provided observational evidence for a late and spatially inhomogeneous end to reionization.

PROFESSIONAL SERVICE

- **Grant Reviewer**: NSF Astronomy and Astrophysics Research Grants Review Panel 2026
- **Peer Reviewer**: *The Astrophysical Journal*, *ApJ Letters*, *JCAP*, and *Open Research Europe*; ALMA and Gemini observing proposals

EDUCATION

Ph.D. in Physics, University of California, Riverside Mar. 2024
Constraining the IGM during the Later Stages of Reionization using QSO Spectra; Advisor: Prof. George Becker

M.Sc. in Physics, University of California, Riverside Dec. 2019

B.Sc. in Astronomy, University of Science and Technology of China Jun. 2018
Testing Gravity Theories on Galactic Scales; Advisor: Prof. Xiao-Bo Dong

EXTERNAL FUNDING & PI PROGRAMS

Funding summary. Four awarded Roman Cycle 1 programs total \$1.20M: \$0.60M from two PI programs provides direct research support for Yongda Zhu, while \$0.60M from two co-PI programs supports collaborators. PI of two ALMA programs and three MMT/Binospec IFU programs; science and observing lead for repeated Keck campaigns and major contributor to JWST GO 4092.

Principal-Investigator Observing Programs

- **Roman 10995** — *Establishing the Post-Reionization Lyα Transmission–Environment Baseline*. **PI: Yongda Zhu**; \$150,000 awarded as direct PI research support.
- **Roman 19049** — *From Environment to Assembly: Roman Spatially Resolved Stellar Mass and SFH Maps*. **PI: Yongda Zhu**; \$450,000 awarded as direct PI research support.
- **Roman 19054** — *Do Filaments Influence Galaxies Beyond Local Environment? A Roman Census of Galaxy–Filament Connectivity*. **co-PI: Yongda Zhu**; \$450,000 awarded to support collaborators.
- **Roman 19029** — *Linking Galaxies to Their Neutral Gas Reservoir: A Roman–Rubin Synergy Survey of DLA Host Galaxies at Cosmic Noon*. **co-PI: Yongda Zhu**; \$150,000 awarded to support collaborators.
- **MMT/Binospec IFU 2026A** — *IFU Observations of a $z = 5.63$ Quasar–Companion System: Lyα Escape and UV Counterpart*. **PI: Yongda Zhu**.
- **MMT/Binospec IFU 2025B** — *Spatially Resolved Lyα Spectroscopy of a $z = 5.5$ Quasar with Strong Cold Outflows*. **PI: Yongda Zhu**.
- **ALMA 2024.1.00555.S (Cycle 11)** — *Galaxy Over/Underdensities Around IGM Transmission at $z = 5.7$: A Robust Constraint on Reionization*. **PI: Yongda Zhu**.
- **MMT/Binospec IFU 2024B** — *Ionization and Enrichment in the Reionization Epoch: A*

Pilot Study with Binospec IFU. **PI: Yongda Zhu**.
- **ALMA 2022.1.00662.S (Cycle 9)** — *The Mean Free Path of Ionizing Photons at $z = 5.6$: A Robust Constraint on Reionization*. **PI: Yongda Zhu**.

Observing Programs Led as a Ph.D. Student at UC Riverside
I led or published the science case and observing program for each allocation below. UC students could not serve as Keck PIs.

- JWST 4092 – *How Does Reionization End? A Search for [O III] Emitters in the Most Transparent Regions of the IGM Near Redshift Six*. Formal PI: George Becker.
  Zhu, et al. 2025, arXiv:2510.09568
- Keck/ESI [2021B_U036, 2022A_U035, 2022B_U042, 2023B_U049, 2024A_U281]: *The Mean Free Path at $z = 5.6$: Insights into Ultra-Late Reionization*.
- Keck/ESI [2019A_U014, 2020A_U121, 2021A_U039]: *Giant Lyα Troughs at $z < 6$: A Signature of Very Late Reionization?*.
  Zhu, et al. 2021, ApJ, 923, 223; **Zhu**, et al. 2022, ApJ, 932, 76
- Keck/LRIS [2019A_U147, 2019B_U147]: *The Mean Free Path at $z = 5$: A Key Constraint on Reionization Models*.
  Becker et al. 2021; **Zhu**, et al. 2023, ApJ, 955, 115
- Other allocations include: Keck/ESI [2021A, 2021B], Keck/DEIMOS [2020A], Keck/LRIS [2023B], Subaru/HSC [2020B, 2021A, 2021B, 2023B]

Selected Awards and Fellowships

- NSF/NRAO Student Observing Support Award ($40,000) 2023
- UCR Dissertation Year Fellowship ($7,200) 2022
- Benjamin C. Shen Memorial Award for Outstanding First-Year Graduate Researcher 2019
- Dean's Distinguished Fellowship, UC Riverside 2018
- Undergraduate research awards and scholarships, USTC 2015–2018
  CAS Undergraduate Research Award, NSFC Talented Students in Basic Sciences Program, CAS Innovation Training Program, and Outstanding Student Scholarships.

SELECTED FIRST-AUTHOR PUBLICATIONS (9 OF 16) & FIELD IMPACT

1. **Zhu, Y.**, et al. 2026, *Low Lyα Visibility in Galaxy Overdensities: Reionization Topology and Neutral-Fraction Ceilings from DIVER over $4.8 < z < 11$*, arXiv:2608.19311.
   Impact: Mapped Lyα visibility for 250 DIVER galaxies over $4.8 < z < 11$, revealing suppressed strong Lyα emission in overdensities—opposite the simplest inside-out reionization expectation—and deriving neutral-fraction ceilings that disfavor an almost completely neutral IGM at $z \sim 8$.
2. **Zhu, Y.**, et al. 2026, *Early metal-enriched baryon cycling before the midpoint of cosmic reionization*, *Nature Astronomy*. doi:10.1038/s41550-026-02988-2.
   Impact: Established metal-enriched baryon cycling in luminous galaxies roughly 500 Myr after the Big Bang through multiphase, blueshifted metal absorption at $7.2 < z < 9.3$.

3. **Zhu, Y.**, et al. 2026, *Galaxy Underdensities Host the Clearest Intergalactic Medium Lyα Transmission and Indicate Anisotropic Reionization*, *ApJ*, 1002, 93.
Impact: Located the two clearest $z \sim 5.7$ IGM-transmission regions in galaxy underdensities below half the cosmic mean and delivered the first two-dimensional galaxy–IGM cross-correlation, revealing anisotropic ionization geometry.
4. **Zhu, Y.**, et al. 2025, *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$*, *ApJL*, 995, L5.
Impact: Provided evidence that quasar radiation suppresses recent star formation out to approximately 7 comoving Mpc, constraining a roughly 3.1 Myr radiative episode and favoring molecular-gas photodissociation.
5. **Zhu, Y.**, et al. 2025, *SMILES: Potentially Higher Ionizing Photon Production Efficiency in Overdense Regions*, *ApJ*, 986, 18.
Impact: Established the first empirical link between environment and ionizing-photon production efficiency, finding a 2.43-fold higher mean $\xi_{\rm ion}$ in overdense regions and tying environment to reionization topology.
6. **Zhu, Y.**, et al. 2025, *A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images*, *ApJ*, 986, 162.
Impact: Created the first systematic JWST medium-band imaging search for extended line emission and potential outflows, identifying 326 candidates across $1.4 < z < 8.4$ and enabling a scalable census of potential feedback systems.
7. **Zhu, Y.**, et al. 2024, *Damping Wing-Like Features in the Stacked Lyα Forest: Potential Neutral Hydrogen Islands at $z < 6$*, *MNRAS*, 533, L49.
Impact: Detected a damping-wing-like signature around Lyβ dark gaps at mean $z = 5.8$, implying $\langle x_{\rm HI} \rangle \geq 6.1 \pm 3.9\%$ and robustly showing that reionization extended below $z = 6$.
8. **Zhu, Y.**, et al. 2023, *Probing Ultralate Reionization: Direct Measurements of the Mean Free Path over $5 < z < 6$*, *ApJ*, 955, 115.
Impact: Delivered the first direct, multi-bin mapping of the ionizing-photon mean free path across $5 < z < 6$, including first measurements at $z \simeq 5.3$ and 5.6, and showed a fully ionized, relaxed IGM was inconsistent with the data as late as $z = 5.3$.
9. **Zhu, Y.**, et al. 2021, *Chasing the Tail of Cosmic Reionization with Dark Gap Statistics in the Lyα Forest over $5 < z < 6$*, *ApJ*, 923, 223.
Impact: Established the long-dark-gap fraction as a new reionization diagnostic, tracing its decline from approximately 90% at $z = 6$ to 15% at $z = 5.6$ and finding neutral-island and/or ionizing-background signatures to $z \simeq 5.3$.

RESEARCH & COLLABORATION LEADERSHIP

- **SAPPHIRES Data Lead**: leads data reduction, validation, and delivery for a large JWST pure-parallel imaging and spectroscopy collaboration.
- **SMILES NIRSpec Lead**: leads NIRSpec data reduction, analysis, and the public high-level science-product release.
- **COSMOS-3D Large-Scale Structure Working Group Co-chair**: coordinates environmental and cosmic-web science across the collaboration.

- **Additional major collaborations**: JWST/JADES, JWST/DIVER, ESO VLT XQR-30, and Roman Science Collaboration.

Major-Program Portfolio

- **JWST:** Co-I or data/analysis lead on approximately 2,006 hr of allocated prime, survey, and pure-parallel time across the programs listed below. The total includes approximately 1,600 hr from SAPPHIRES and LAPIS and approximately 406 hr from targeted GO programs.
- **ALMA:** PI of two programs and Co-I on the six programs listed below.
- **ESO/VLT:** Member of Large Programme 112.262L (PI: Emanuele Paolo Farina), *The Cosmic Ecosystem of the First QSOs and Galaxies: a MUSE/XSHOOTER/JWST/ALMA Legacy Survey*; 66 hr with MUSE.

Additional ALMA Co-Investigator Programs

- 2025.1.01067.S (PI: Zhiyuan Ji): *Dust Rising from the ASHES: A Comprehensive ALMA Survey of Remnant Dust in the Earliest Quiescent Galaxies*.
- 2025.1.00805.S (PI: Weizhe Liu): *The Molecular Phase of the Earliest Galaxy-scale Outflows in Quasars at* $z > 6.5$.
- 2025.1.00147.S (PI: Joris Witstok): *Reionization Revealed: ALMA Anatomizes an Enigmatic Ionizing Source within a Galaxy at Redshift 13*.
- 2025.1.00493.S (PI: Manuela Bischetti): *Investigating the Physical Properties and Origin of an Extended [C II] Halo at* $z \sim 6$.
- 2025.1.00215.S (PI: Fengwu Sun): *Dust-Obscured Starbursts in the Most Massive Dark Matter Halos at* $z = 8$*–9: An ALMA+JWST Wide-Field Census*.
- 2024.1.00469.S (PI: Manuela Bischetti): *Investigating the Physical Properties and Origin of an Extended [C II] Halo at* $z \sim 6$.

Selected Major Co-Investigator Programs:

- JWST 12583 – The Dragon Survey: A Direct Probe of the Early Stellar Luminosity Function and Dark Matter through Multi-cycle JWST Multi-cadence Microlensing at $z = 0.73$
- JWST 12577 – JWST Multi-Cycle Deep Transient Survey in GOODS-S
- JWST 10790 – LAPIS: Large Area Parallel Imaging Survey
- JWST 10311 – Slicing the Saguaro: Dissecting a Little Red Dot Descendant at Cosmic Noon with NIRSpec/IFU
- JWST 8544 – Rest-frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-up for JADES-GS-z14-0
- JWST 8060 – JWST Multi-Cycle Deep Transient Survey in GOODS-S
- JWST 8018 – DIVER: Deep Insights into UV Spectroscopy at the Epoch of Reionization
- JWST 7935 – Efficient Measurement of the Emergence Rate of AGN in Legacy Deep Fields
- JWST 7436 – The Last Neutral Islands at the End of Reionization? Characterizing the Nature of the Longest Dark Gaps in IGM Transmission at $z \sim 5.3$
- JWST 7390 – Probing Pair-Instability Supernovae through the Triply-lensed MACS0647-JD at $z = 10.17$

- JWST 7345 – The Dragon Survey: A Direct Probe of the Early Stellar Luminosity Function and Dark Matter through Multi-cycle JWST Multi-cadence Microlensing at $z = 0.73$
- JWST 7336 – Commission a new $R \sim 2500$ "IFU" on JWST: Calibrating second-order spectra of NIRCam/WFSS through the Hubble Ultra Deep Field
- JWST 7335 – Forever Blowing Bubbles: What Powers a 24-kpc Ionized Gas Nebula Around a Normal Galaxy at $z = 6$?

TEACHING & MENTORING

- **Teaching Assistant**, UC Riverside
  - **Summer PhD Comprehensive-Exam Review Course** 2023
    Taught 30 Physics PhD students in an intensive course covering classical mechanics, electromagnetism, thermodynamics, and statistical mechanics.
  - **Undergraduate General Physics Laboratories** 2018–2019
    Led PHYS 2LA (Physics Lab A; 40 students) and PHYS 2LB (Physics Lab B; 40 students); teaching evaluations ranked in the top 20% across campus.
- **Teaching Assistant**, USTC
  - **Introductory C/C++ Programming** 2016
    Taught data structures and algorithms to 160 enrolled students.
- **Zheng Ma**, University of Arizona graduate student 2025–present
  - **Research project:** High-redshift AGN and host-galaxy evolution.
  - **First-author paper:** *A quasar hatching from a buried red phase at* $z = 3.7$ (arXiv:2607.22966).
  - **First-author paper:** *Undermassive Hosts of* $z = 4$*–6 AGN from JWST/NIRCam Image Decomposition* (arXiv:2601.15962; submitted to *ApJ*).
- **Suprabhas Narisetty**, University of Arizona undergraduate 2025–present
  - **Research project:** JWST/NIRSpec study of cosmic-noon galaxies.
  - **Outcome:** First-author manuscript *Strong Balmer Breaks Mark Gradually Quenching Galaxies Below the Star-forming Main Sequence at* $1 < z < 4$ *with JWST/NIRSpec*, prepared for *ApJ*; DOI: 10.5281/zenodo.17772416.
- **G. Hernandez**, UC Riverside undergraduate 2021
  - **Research project:** Evolution of intergalactic-medium effective optical depth.
- **Seyedazim Hashemi**, UC Riverside PhD student — mentored as a peer
  - **Research topic:** Lyα visibility during reionization.
  - **Outcome:** First-author paper *Lyα emission from [O III] emitters near reionization: the role of environment in galaxy Lyα detection*, *MNRAS*, 542, 104 (2025).
- **UCR International Students & Scholars Office** 2021–2023
  - Career and academic mentoring for 14 international graduate students.
- **UCR Graduate Student Mentorship Program** 2020
  - Mentored two junior PhD students across STEM: N. Ahvazi (dark matter and galaxies) and Q. Wu (2D materials).

Courses prepared to teach: introductory astronomy and physics; galaxies and cosmology; observational astronomy; astronomical techniques; computational astrophysics; scientific programming;

and astronomical data analysis.

SELECTED INVITED TALKS & CONFERENCE PRESENTATIONS

* invited

- * Star-forming Galaxies in the Reionization Epoch, Ishigaki, Japan — March 2026
- Galaxy Origins in the JWST Era, Toledo — May 2025
- JWST NIRCam Science Meeting, Biosphere 2 — March 2025
- * NOIRLab Joint Colloquium, **University of Arizona** — February 2025
- * Cosmology Seminar, **Arizona State University** — February 2025
- Lyman-alpha Forest Workshop, **Ohio State University** — October 2024
- JWST MIRI Science Meeting, Biosphere 2 — October 2024
- The First Gigayear(s), Hilo, Hawai‘i — September 2024
- JADES Collaboration Meeting, **University of Copenhagen** — April 2024
- * Galaxy Group Talk, **University of Arizona** — March 2024
- * Galaxy Seminar, **University of Michigan** — November 2023
- Galaxy Formation and Evolution in Southern California — September 2023
- * Kashiwa-Mitaka Meeting (KMM) Seminar, **University of Tokyo** — August 2023
- * Lightning Talk, First Light Conference, **MIT** — June 2023
- Reionisation in the Summer Conference, **MPIA**, Heidelberg — June 2023
- * Galaxy Formation Group, **Northwestern University (CIERA)** — December 2022
- * FLASH Seminar, **UC Santa Cruz** — November 2022
- * Astronomy Lunch Talk, **UC Los Angeles** — October 2022
- * Astro Lunch Talk, **UC Santa Barbara** — September 2022
- Reionization on a Blackboard Workshop, **CCA** — September 2022
- * Special arXiv Coffee, **UC Davis** — May 2022
- * Physics & Astronomy Seminar, **UC Riverside** — April 2022
- * High-z Group Talk, **Tsinghua University** — April 2022
- Reionization and Cosmic Dawn, **UC Berkeley** — March 2022
- European Astronomical Society Annual Meeting (EAS 2021) — July 2021
- Summer All Zoom Epoch of Reionization Conference — June 2021
- * EURECA Seminar, **University of Arizona** — February 2021

PUBLIC OUTREACH & SERVICE

- Invited public talk at Arizona Senior Academy — April 2026
- Session Chair, AAS 247 — January 2026
- Chambliss Judge for student poster competition, AAS 247 — January 2026
- Instructor, *Galactic Adventures*, Flandrau Science Center & Planetarium — June 2025
- Stargazing outreach event, Home Gardens Library (Corona, CA) — October 2023
- Co-organizer, UCR Physics & Astronomy Student Seminar — 2022–2023
- Instructor, UCR Camp Highlander — Summer 2022
- Designed and taught outreach courses for K–12 students — 2022

  - *Multiwavelength Universe*, *Gravity Simulator*
- Virtual Stargazing (UCR & Riverside Astronomical Society): monthly live public outreach on YouTube 2020–2021
- Science fair judge, Riverside County Science and Engineering Fair 2021–2023
- UCR Astronomy Public Outreach: Mercury Transit November 2019

OPEN SCIENCE & COMMUNITY RESOURCES

- **SMILES NIRSpec HLSP (JWST-GTO-1207)**: Lead developer of the public high-level data release including 1D/2D spectra and derived galaxy catalogs.
  GitHub: github.com/ydzhuastro/smiles_dr2; MAST HLSP: doi:10.17909/et3f-zd57
- **Keck/ESI Rest-UV Spectra for Bright Quasars**: Reduced and calibrated spectra for $m_z < 21$ quasars at $z \sim 5.5$, publicly available on Zenodo.
  doi.org/10.5281/zenodo.18677315.
- **Tutorial: Galaxy Morphology × Machine Learning** February 2025
  Introducing morphology classification and neural network tools for students.
  EURECA Tutorial PDF

SELECTED MEDIA COVERAGE

- "Galassie al lavoro nell'universo bambino" — **Media INAF** Oct. 2026
- "James Webb Space Telescope looks back in time, finds our ancient universe wasn't as pure as we thought" — **Space.com** Sept. 2026
- "JWST finds early galaxies were already seeding the universe with heavy elements" — **University of Arizona News** Sept. 2026
- "Supermassive serial killers: Astronomers discover how black holes 'kill off' neighboring galaxies" — **Space.com** Feb. 2026
- "Cosmic predators: How supermassive black holes slow star growth in nearby galaxies" — **University of Arizona News** Feb. 2026
- "Should I Stay or Should I Go? Clumps and Inward Migration" — **Astrobites** Feb. 2026
- "Intergalactic Impacts of Quasars During the Epoch of Reionization" — **AAS Nova** Dec. 2025
- "Astronomers discover a unique quasi-stellar object–dusty star-forming galaxy system" — **Phys.org** Nov. 2024
- "Can You Explain These Long, Dark Gaps in Your Cosmological Resume?" — **AAS Nova** Jul. 2022
- "The End of the Cosmic Dawn: Settling a Two-Decade Debate" — **SciTechDaily** Jul. 2022

TECHNICAL EXPERTISE

- **Observational data and pipelines:** JWST/NIRSpec MSA reduction and calibration; NIRCam imaging and WFSS; PSF matching; slitless-spectroscopy analysis; Keck/ESI and LRIS spectroscopy; ALMA/CASA analysis; public high-level data products.
- **Inference and scientific computing:** Bayesian SED fitting; cosmological simulations; spa-

tial and environmental statistics; machine learning and neural networks; high-performance computing.
- **Programming:** Python, C, Julia, IDL/GDL, and working experience with C++, CUDA, CASA, Gadget-4, GALAXY, and FreeFem++.

YONGDA ZHU - PUBLICATION LIST

ORCID: 0000-0003-3307-7525
ADS link: https://ui.adsabs.harvard.edu/search/q=orcid%3A0000-0003-3307-7525
Google Scholar: https://scholar.google.com/citations?user=wDrSZWYAAAAJ

Citation Metrics (as of June 2026):
Total citations: 3500+ | h-index: 35 | First-author citations: 430+ | First-author papers: 16

First-author:

16. **Zhu, Y.**, Fan, X., Keating, L. C., Becker, G. D., Egami, E., Lin, X., Sun, F., Cain, C., Rieke, M. J., Bunker, A. J., Cai, S., D'Eugenio, F., Helton, J. M., Jin, X., Li, M., Ma, Z., Maiolino, R., Rinaldi, P., Willmer, C. N. A., Wu, Y., Wu, Z., and Zhang, J., **2026**. *Low Lyα Visibility in Galaxy Overdensities: Reionization Topology and Neutral-Fraction Ceilings from DIVER over* $4.8 < z < 11$, arXiv e-prints, arXiv:2608.19311. Submitted to *ApJ*.

15. **Zhu, Y.**, Rieke, M. J., Carreira, C., Sun, Y., Ji, Z., Lyu, J., Williams, C. C., Alberts, S., Yuan, F., Zou, Y., Nakazato, Y., Bunker, A. J., Tacchella, S., Sun, F., Rieke, G. H., Egami, E., Chevallard, J., Hainline, K., Ma, Z., Pérez-González, P. G., Rinaldi, P., Rodríguez Del Pino, B., Shields, T., Stone, M., Willmer, C. N. A., Wu, Z., Yue, M., and Zhang, J., **2026**. *PhotoIFU: NIRCam as a Photometric Integral Field Unit for Mapping Feedback in Galaxies*, arXiv e-prints, arXiv:2607.23020.

14. **Zhu, Y.**, Ji, Z., Becker, G. D., Ding, J., Egami, E., Fan, X., Jin, X., Liu, W., Lyu, J., Ma, Z., Narisetty, S., Rieke, G. H., Wu, Y., Yue, M., Zhang, J., and Rieke, M. J., **2026**. *Early metal-enriched baryon cycling before the midpoint of cosmic reionization*, *Nature Astronomy*. doi:10.1038/s41550-026-02988-2.

13. **Zhu, Y.**, Becker, G. D., D'Aloisio, A., Endsley, R., Gangolli, N., Cain, C., Mason, C. A., Hashemi, S., and Hong, H., **2026**. *Galaxy Underdensities Host the Clearest Intergalactic Medium Lyα Transmission and Indicate Anisotropic Reionization*, The Astrophysical Journal, 1002, 93

12. **Zhu, Y.**, Rieke, M. J., Ho, L. C., Sun, Y., Rieke, G. H., Yuan, F., Bakx, T. J. L. C., Becker, G. D., Yang, J., Bañados, E., Bischetti, M., Cain, C., Fan, X., Fudamoto, Y., Hashemi, S., Ikeda, R., Ji, Z., Jin, X., Liu, W., Liu, Y., Lyu, J., Ma, H.-X., Takeuchi, T. T., Umehata, H., Wang, F., and Tee, W. L., **2026**. *A Potential Link between Nuclear Winds and Cold Gas Outflows on Kiloparsec Scales in Reionization-era Quasars*, The Astrophysical Journal, 1000, 312.

11. **Zhu, Y.**, Rieke, M. J., Ji, Z., Bunker, A. J., Carreira, C., Danhaive, A. L., Duan, Q., Egami, E., Eisenstein, D. J., Hainline, K., Johnson, B. D., Ma, Z., Puskás, D., Rieke, G. H., Rinaldi, P., Robertson, B., Tacchella, S., Übler, H., Villanueva, N. C., Williams, C. C., Willmer, C.

N. A., Wu, Z., and Zhang, J., **2026**. *Clumps in High-redshift Galaxies: Mass Scaling and Radial Trends from JADES*, The Astrophysical Journal, 1000, 303.

10. **Zhu, Y.**, Bonaventura, N., Sun, Y., Rieke, G. H., Alberts, S., Lyu, J., Shivaei, I., Morrison, J., Ji, Z., Egami, E., Helton, J. M., Rieke, M. J., Rinaldi, P., Sun, F., and Willmer, C. N. A., **2026**. *SMILES Data Release. II. Probing Galaxy Evolution during Cosmic Noon and Beyond with NIRSpec Medium-resolution Spectra*, The Astrophysical Journal, 997, 301.

9. **Zhu, Y.**, Egami, E., Fan, X., Sun, F., Becker, G. D., Cain, C., Chen, H., Eilers, A.-C., Fudamoto, Y., Helton, J. M., Jin, X., Pudoka, M., Bunker, A. J., Cai, Z., Champagne, J. B., Ji, Z., Lin, X., Liu, W., Ma, H.-X., Ma, Z., Maiolino, R., Rieke, G. H., Rieke, M. J., Rinaldi, P., Sun, Y., Tee, W. L., Wang, F., Yang, J., Yue, M., and Zhang, J., **2025**. *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$*, The Astrophysical Journal, 995, L5.

8. **Zhu, Y.**, Rieke, M. J., Ji, Z., Simmonds, C., Sun, F., Sun, Y., Alberts, S., Bhatawdekar, R., Bunker, A. J., Cargile, P. A., Carniani, S., de Graaff, A., Hainline, K., Helton, J. M., Jones, G. C., Lyu, J., Rieke, G. H., Rinaldi, P., Robertson, B., Scholtz, J., Übler, H., Williams, C. C., and Willmer, C. N. A., **2025**. *A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images*, The Astrophysical Journal, 986, 162.

7. **Zhu, Y.**, Alberts, S., Lyu, J., Morrison, J., Rieke, G. H., Sun, Y., Helton, J. M., Ji, Z., Bhatawdekar, R., Bonaventura, N., Bunker, A. J., Lin, X., Rieke, M. J., Rinaldi, P., Shivaei, I., Willmer, C. N. A., and Zhang, J., **2025**. *SMILES: Potentially Higher Ionizing Photon Production Efficiency in Overdense Regions*, The Astrophysical Journal, 986, 18.

6. **Zhu, Y.**, Bakx, T. J. L. C., Ikeda, R., Umehata, H., Becker, G. D., Cain, C., Champagne, J. B., Fan, X., Fudamoto, Y., Jin, X., Ma, H.-X., Sun, Y., Takeuchi, T. T., and Tee, W. L., **2024**. *Discovery of a Unique Close Quasar–DSFG Pair Linked by a [C II] Bridge at $z = 5.63$*, Research Notes of the American Astronomical Society, 8, 284.

5. **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Cain, C., Keating, L. C., Nasir, F., D'Odorico, V., Bañados, E., Bian, F., Bischetti, M., Bolton, J. S., Chen, H., D'Aloisio, A., Davies, F. B., Davies, R. L., Eilers, A.-C., Fan, X., Gaikwad, P., Greig, B., Haehnelt, M. G., Kulkarni, G., Lai, S., Puchwein, E., Qin, Y., Ryan-Weber, E. V., Satyavolu, S., Spina, B., Walter, F., Wang, F., Wolfson, M., and Yang, J., **2024**. *Damping wing-like features in the stacked Ly α forest: Potential neutral hydrogen islands at $z < 6$*, Monthly Notices of the Royal Astronomical Society, 533, L49.

4. **Zhu, Y.**, Becker, G. D., Christenson, H. M., D'Aloisio, A., Bosman, S. E. I., Bakx, T., D'Odorico, V., Bischetti, M., Cain, C., Davies, F. B., Davies, R. L., Eilers, A.-C., Fan, X., Gaikwad, P., Haehnelt, M. G., Keating, L. C., Kulkarni, G., Lai, S., Ma, H.-X., Mesinger, A., Qin, Y., Satyavolu, S., Takeuchi, T. T., Umehata, H., and Yang, J., **2023**. *Probing Ultralate Reionization: Direct Measurements of the Mean Free Path over $5 < z < 6$*, The Astrophysical Journal, 955, 115.

3. **Zhu, Y.**, Ma, H.-X., Dong, X.-B., Huang, Y., Mistele, T., Peng, B., Long, Q., Wang, T., Chang, L., and Jin, X., **2023**. *How close dark matter haloes and MOND are to each other:*

three-dimensional tests based on Gaia DR2, Monthly Notices of the Royal Astronomical Society, 519, 4479.

2. **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Keating, L. C., D'Odorico, V., Davies, R. L., Christenson, H. M., Bañados, E., Bian, F., Bischetti, M., Chen, H., Davies, F. B., Eilers, A.-C., Fan, X., Gaikwad, P., Greig, B., Haehnelt, M. G., Kulkarni, G., Lai, S., Pallottini, A., Qin, Y., Ryan-Weber, E. V., Walter, F., Wang, F., and Yang, J., **2022**. *Long Dark Gaps in the Lyβ Forest at $z < 6$: Evidence of Ultra-late Reionization from XQR-30 Spectra*, The Astrophysical Journal, 932, 76.

1. **Zhu, Y.**, Becker, G. D., Bosman, S. E. I., Keating, L. C., Christenson, H. M., Bañados, E., Bian, F., Davies, F. B., D'Odorico, V., Eilers, A.-C., Fan, X., Haehnelt, M. G., Kulkarni, G., Pallottini, A., Qin, Y., Wang, F., and Yang, J., **2021**. *Chasing the Tail of Cosmic Reionization with Dark Gap Statistics in the Lyα Forest over* $5 < z < 6$, The Astrophysical Journal, 923, 223.

Student-led papers:

3. Ma, Z. (UofA grad student), **Zhu, Y.**, Ji, Z., Egami, E., Rieke, M. J., Fan, X., Lyu, J., Rieke, G. H., Sun, F., Sun, Y., Becker, G. D., Bunker, A. J., D'Eugenio, F., Jin, X., Juodžbalis, I., Liu, W., Maiolino, R., Rinaldi, P., Wang, F., Willmer, C. N. A., Wu, Y., Yang, J., Zhang, J., and Zhu, P., **2026**. *A quasar hatching from a buried red phase at z = 3.7*, arXiv e-prints, arXiv:2607.22966.

2. Ma, Z. (UofA grad student), Egami, E., **Zhu, Y.**, Sun, F., Lyu, J., Zhang, J., Willmer, C. N. A., Bunker, A. J., Carniani, S., Curtis-Lake, E., Hausen, R., Ji, X., Ji, Z., Juodžbalis, I., Maiolino, R., Rieke, G. H., Rinaldi, P., Sun, Y., Tacchella, S., Übler, H., and Williams, C. C., **2026**. *Undermassive Hosts of $z = 4 - 6$ AGN from JWST/NIRCam Image Decomposition with CONGRESS, FRESCO, and JADES*, arXiv e-prints, arXiv:2601.15962. Submitted to ApJ.

1. Narisetty, S. (UofA undergraduate), **Zhu, Y.**, and Egami, E., **2025**. *Strong Balmer Breaks Mark Gradually Quenching Galaxies Below the Star-forming Main Sequence at* $1 < z < 4$ *with JWST/NIRSpec*, for ApJ submission. DOI: 10.5281/zenodo.17772416

Co-author:

83. Sun, Y., Ji, Z., D'Eugenio, F., **Zhu, Y.**, Rieke, G. H., Baker, W. M., Bunker, A. J., Carniani, S., Helton, J. M., Perna, M., Pérez-González, P. G., Rinaldi, P., Übler, H., and Willmer, C. N. A., **2026**. *A Census of Na D-traced neutral ISM and outflows at* $0.6 < z < 4$, arXiv e-prints, arXiv:2604.18522.

82. Lebowitz, S., Hainline, K. N., Juneau, S., Williams, C. C., Panda, S., Lyu, J., Maseda, M. V., Tacchella, S., **Zhu, Y.**, and Aguayo, J. L., **2026**. *Spatially Resolved AGN Ionization and Star Formation at Cosmic Noon with JWST/JEMS*, arXiv e-prints, arXiv:2604.01271.

81. Danhaive, A. L., Tacchella, S., McClymont, W., Robertson, B., Carniani, S., Carreira, C., Egami, E., Bunker, A. J., Curtis-Lake, E., Eisenstein, D. J., Ji, Z., Johnson, B. D., Rieke, M., Villanueva, N. C., Willmer, C. N. A., Willott, C., Wu, Z., and **Zhu, Y.**, **2026**. *Beyond*

the stars: Linking Hα sizes, kinematics, and star formation in galaxies at $z \approx 4-6$ with JWST grism surveys and GEKO, Monthly Notices of the Royal Astronomical Society, 547, stag437.

80. Danhaive, A. L., Tacchella, S., Bunker, A. J., Curtis-Lake, E., de Graaff, A., D'Eugenio, F., Duan, Q., Egami, E., Eisenstein, D. J., Johnson, B. D., Maiolino, R., McClymont, W., Rieke, M., Robertson, B., Sun, F., Willmer, C. N. A., Wu, Z., and **Zhu, Y.**, **2026**. *The dark side of early galaxies: GEKO uncovers dark-matter fractions at $z \sim 4-6$*, Monthly Notices of the Royal Astronomical Society, 546, stag119.

79. Rowlands, S., Davies, R. L., Ryan-Weber, E., Keating, L. C., Sebastian, A. M., Becker, G. D., Bischetti, M., Bosman, S. E. I., Chen, H., Davies, F. B., D'Odorico, V., Gaikwad, P., Gallerani, S., Haehnelt, M. G., Kulkarni, G., Meyer, R. A., Welsh, L., and **Zhu, Y.**, **2026**. *E-XQR-30: evidence for an increase in the ionization state of metal absorbers from $z \sim 6$ to $z \sim 2$*, Monthly Notices of the Royal Astronomical Society, 546, staf2254.

78. Lin, X., Fan, X., Cai, Z., Bian, F., Liu, H., Sun, F., Ma, Y., Greene, J. E., Strauss, M. A., Green, R., Lyu, J., Champagne, J. B., Goulding, A. D., Inayoshi, K., Jin, X., Leung, G. C. K., Li, M., Liu, W., Liu, Y., Mao, J., Pudoka, M. A., Tee, W. L., Wang, B., Wang, F., Wu, Y., Yang, J., Zhang, H., and **Zhu, Y.**, **2026**. *The Discovery of Little Red Dots in the Local Universe: Signatures of Cool Gas Envelopes*, The Astrophysical Journal, 997, 364.

77. Zhang, J., Egami, E., Sun, F., Lin, X., Lyu, J., **Zhu, Y.**, Rinaldi, P., Sun, Y., Bunker, A. J., Bhatawdekar, R., Helton, J. M., Maiolino, R., Ma, Z., Robertson, B., Tacchella, S., Venturi, G., Williams, C. C., and Willott, C., **2026**. *Abundant Population of Broad Hα Emitters in the GOODS-N Field Revealed by CONGRESS, FRESCO, and JADES*, The Astrophysical Journal, 997, 250.

76. Lin, X., Egami, E., Sun, F., Zhang, H., Fan, X., Helton, J. M., Wang, F., Bunker, A. J., Cai, Z., Eisenstein, D. J., Jaffe, D. T., Ji, Z., Jin, X., Pudoka, M. A., Tacchella, S., Tee, W. L., Rinaldi, P., Robertson, B., Sun, Y., Willmer, C. N. A., Willott, C., Zhang, J., and **Zhu, Y.**, **2026**. *The Luminosity Function and Clustering of Hα Emitting Galaxies at $z \asymp 4-6$ from a Complete NIRCam Grism Redshift Survey*, The Astrophysical Journal, 997, 207.

75. Sun, Y., Ji, Z., Rieke, G. H., D'Eugenio, F., **Zhu, Y.**, Sun, F., Lin, X., Bunker, A. J., Lyu, J., Rinaldi, P., and Willmer, C. N. A., **2026**. *Extreme Neutral Outflow in a Non-active Galactic Nucleus Quiescent Galaxy at $z \sim 1.3$*, The Astrophysical Journal, 997, 140.

74. Sun, F., Eisenstein, D. J., D'Eugenio, F., Hainline, K., Helton, J. M., Johnson, B. D., Lin, X., Rieke, M., Robertson, B., Tacchella, S., Bunker, A. J., Chevallard, J., Curtis-Lake, E., Egami, E., Hausen, R., Ji, Z., Lyu, J., Maiolino, R., Rinaldi, P., Sun, Y., Trussler, J. A. A., Williams, C. C., Willmer, C. N. A., Witstok, J., Wu, Z., and **Zhu, Y.**, **2026**. *JADES: Discovery of Large Reservoirs of Small Dust Grains in the Circumgalactic Medium of Massive Galaxies at $z \sim 3.5$ through Deep JWST/NIRCam Imaging and Grism Spectroscopy*, arXiv e-prints, arXiv:2601.15961.

73. Carreira, C., Robertson, B. E., Danhaive, A. L., Ji, Z., Rieke, M., Tacchella, S., Villanueva, N. C., Willmer, C. N. A., Wu, Z., **Zhu, Y.**, Baker, W. M., Bunker, A. J., Cameron, A. J., Chevallard, J., Curtis-Lake, E., Duan, Q., Eisenstein, D. J., Hainline, K., Hausen, R.,

Johnson, B. D., Maiolino, R., Mengistu, P., Puskás, D., Rinaldi, P., Sun, Y., Trussler, J. A. A., Übler, H., Uppal, A., and Williams, C. C., **2026**. *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Catalogs of inferred morphological properties of galaxies from JWST/NIRCam imaging in GOODS-N and GOODS-S*, arXiv e-prints, arXiv:2601.15957.

72. Robertson, B. E., Johnson, B. D., Tacchella, S., Eisenstein, D. J., Hainline, K., Alberts, S., Arribas, S., Baker, W. M., Bunker, A. J., Cameron, A. J., Carniani, S., Carreira, C., Chevallard, J., Circosta, C., Curtis-Lake, E., Danhaive, A. L., Duan, Q., Egami, E., Hausen, R., Helton, J. M., Ji, Z., Maiolino, R., Pérez-González, P. G., Puskás, D., Rieke, M., Rinaldi, P., Sun, F., Sun, Y., Übler, H., Trussler, J. A. A., Villanueva, N. C., Whitler, L., Williams, C. C., Willmer, C. N. A., Willott, C., Wu, Z., and **Zhu, Y.**, **2026**. *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometric Catalog*, arXiv e-prints, arXiv:2601.15956.

71. Johnson, B. D., Robertson, B. E., Eisenstein, D. J., Tacchella, S., Puskás, D., Duan, Q., Wu, Z., Hainline, K., Rieke, M., Willott, C., Willmer, C. N. A., Trussler, J. A. A., Alberts, S., Arribas, S., Baker, W. M., Bunker, A. J., Cameron, A. J., Carniani, S., Carreira, C., Cargile, P. A., Curtis-Lake, E., Egami, E., Hausen, R., Helton, J. M., Ji, Z., Maiolino, R., Pérez-González, P. G., Rinaldi, P., Sun, F., Sun, Y., Villanueva, N. C., Williams, C. C., and **Zhu, Y.**, **2026**. *JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: NIRCam Imaging in GOODS-S and GOODS-N*, arXiv e-prints, arXiv:2601.15954.

70. Davies, F. B., Bosman, S. E. I., D'Odorico, V., Campo, S., Mesinger, A., Qin, Y., Becker, G. D., Bañados, E., Chen, H., Cristiani, S., Fan, X., Gallerani, S., Haehnelt, M. G., Keating, L. C., Lai, S., Ryan-Weber, E., Wang, F., Yang, J., and **Zhu, Y.**, **2026**. *Updated dark pixel fraction constraints on reionization's end from the Lyman-series forests of XQR−30*, Monthly Notices of the Royal Astronomical Society, 545, staf1862.

69. Jin, X., Wang, F., Yang, J., Fan, X., Bian, F., Li, J.-T., Liu, W., Liu, Y., Lyu, J., Pudoka, M., Tee, W. L., Wu, Y., Zhang, H., and **Zhu, Y.**, **2026**. *A Deep Chandra X-Ray Survey of a Luminous Quasar Sample at $z \sim 7$*, The Astrophysical Journal, 997, 83.

68. Lin, X., Fan, X., Sun, F., Zhang, J., Egami, E., Helton, J. M., Wang, F., Zhang, H., Bunker, A. J., Cai, Z., Ji, Z., Jin, X., Maiolino, R., Pudoka, M. A., Rinaldi, P., Robertson, B., Tacchella, S., Tee, W. L., Sun, Y., Willmer, C. N. A., Willott, C., and **Zhu, Y.**, **2026**. *The Large-scale Environments of Low-luminosity AGNs at $3.9 < z < 6$ and Implications for Their Host Dark Matter Halos from a Complete NIRCam Grism Redshift Survey*, The Astrophysical Journal, 997, 61.

67. Helton, J. M., Morrison, J. E., Hainline, K. N., D'Eugenio, F., Rieke, G. H., Alberts, S., Carniani, S., Leja, J., Li, Y., Rinaldi, P., Scholtz, J., Stone, M., Willmer, C. N. A., Wu, Z., Baker, W. M., Bunker, A. J., Charlot, S., Chevallard, J., Cleri, N. J., Curti, M., Curtis-Lake, E., Egami, E., Eisenstein, D. J., Jakobsen, P., Ji, Z., Johnson, B. D., Kumari, N., Lin, X., Lyu, J., Maiolino, R., Maseda, M., Pérez-González, P. G., Rieke, M. J., Robertson, B., Saxena, A., Sun, F., Tacchella, S., Übler, H., Venturi, G., Williams, C. C., Willott, C., Witstok, J., and **Zhu, Y.**, **2025**. *Ionizing Photon Production Efficiencies and Chemical Abundances at Cosmic Dawn Revealed by Ultra-Deep Rest-Frame Optical Spectroscopy of JADES-GS-z14-0*, arXiv e-prints, arXiv:2512.19695.

66. Liu, W., Fan, X., Li, H., Green, R., Champagne, J. B., Jin, X., Lyu, J., Pudoka, M., Tee, W. L., Wang, F., Yang, J., **Zhu, Y.**, and Abdessalam, N., **2025**. *A JWST/NIRSpec Integral Field Unit Survey of Luminous Quasars at z 5-6 (Q-IFU): Rest-frame Optical Nuclear Properties and Extended Nebulae*, arXiv e-prints, arXiv:2511.06085.

65. Rinaldi, P., Pérez-González, P. G., Rieke, G. H., Lyu, J., D'Eugenio, F., Wu, Z., Carniani, S., Looser, T. J., Shivaei, I., Boogaard, L. A., Diaz-Santos, T., Colina, L., Östlin, G., Alberts, S., Álvarez-Márquez, J., Annuziatella, M., Aravena, M., Bhatawdekar, R., Bunker, A. J., Caputi, K. I., Charlot, S., Crespo Gómez, A., Curti, M., Eckart, A., Gillman, S., Hainline, K., Kumari, N., Hjorth, J., Iani, E., Inami, H., Ji, Z., Johnson, B. D., Jones, G. C., Labiano, Á., Maiolino, R., Melinder, J., Moutard, T., Peissker, F., Rieke, M., Robertson, B., Scholtz, J., Tacchella, S., van der Werf, P. P., Walter, F., Williams, C. C., Willott, C., Witstok, J., Übler, H., and **Zhu, Y.**, **2025**. *Deciphering the Nature of Virgil: An Obscured Active Galactic Nucleus Lurking within an Apparently Normal Lyα Emitter during Cosmic Reionization*, The Astrophysical Journal, 994, 86.

64. Rieke, G. H., Sun, Y., Lyu, J., Willmer, C. N. A., **Zhu, Y.**, Rinaldi, P., Stone, M. A., Hainline, K. N., and Pérez-González, P. G., **2025**. *Confirming Near- to Mid-infrared Photometrically Identified Obscured AGNs in the JWST Era*, The Astrophysical Journal, 994, 35.

63. Stone, M. A., Rieke, G. H., Lyu, J., Florian, M. K., Hainline, K. N., Sun, Y., and **Zhu, Y.**, **2025**. *The z = 7.08 Quasar ULAS J1120+0641 May Never Reach a "Normal" Black Hole to Stellar Mass Ratio*, The Astrophysical Journal, 993, 168.

62. Welsh, L., D'Odorico, V., Fontanot, F., Davies, R., Bosman, S. E. I., Cupani, G., Becker, G., Keating, L., Ryan-Weber, E., Bischetti, M., Haehnelt, M., Chen, H., **Zhu, Y.**, Lai, S., Hirschmann, M., Xie, L., and Qin, Y., **2025**. *The clustering of C IV and Si IV at the end of reionization: A perspective from the E-XQR-30 survey*, Astronomy and Astrophysics, 703, A274.

61. Fudamoto, Y., Nakazato, Y., Ceverino, D., Colina, L., Hashimoto, T., Inoue, A. K., Tamura, Y., Yoshida, N., **Zhu, Y.**, Sugahara, Y., Arribas, S., 'Arvarez-M'arquez, J., Bakx, T., Blanco Prieto, C., Costantin, L., Crespo G'omez, A., Hagimoto, M., Hashigaya, T., Matsuo, H., Marques-Chaves, R., Mawatari, K., Mitsuhashi, I., Osone, W., Pereira-Santaella, M., Umehata, H., Witten, C., and Ren, Y. W., **2025**. *Early massive galaxy formation in the core of a galaxy protocluster 650 million years after the Big Bang*, arXiv e-prints, arXiv:2510.11770.

60. D'Eugenio, F., Nelson, E. J., Eisenstein, D. J., Maiolino, R., Carniani, S., Scholtz, J., Curti, M., Willmer, C. N. A., Bunker, A. J., Helton, J. M., Juodžbalis, I., Sun, F., Tacchella, S., Arribas, S., Cameron, A. J., Charlot, S., Curtis-Lake, E., Hainline, K., Johnson, B. D., Robertson, B., Williams, C. C., Willott, C., Baker, W. M., Chevallard, J., Danhaive, A. L., Isobe, Y., Ji, X., Ji, Z., Jones, G. C., Kumari, N., Looser, T. J., Lyu, J., Parlanti, E., Perna, M., Puskás, D., Rinaldi, P., Simmonds, C., Sun, Y., Übler, H., Venturi, G., Witstok, J., Wu, Z., and **Zhu, Y.**, **2025**. *JADES Dark Horse: demonstrating high-multiplex observations with JWST/NIRSpec dense-shutter spectroscopy in the JADES Origins Field*, arXiv e-prints, arXiv:2510.11626.

59. Meyer, R. A., Wang, F., Kakiichi, K., Brammer, G., Champagne, J., Jurk, K., Li, Z., Li, Z.,

Musin, M., Satyavolu, S., Schindler, J.-T., Shuntov, M., Xu, Y., Zou, S., Bian, F., Casey, C., Egami, E., Fan, X., Jiang, D., Laporte, N., Liu, W., Oesch, P., Tasca, L., Yang, J., Zhang, Z., Akins, H., Cai, Z., Coulter, D. A., Huang, J., Li, M., Liu, W., Liang, Y., Jin, X., Kartaltepe, J., Matharu, J., Pudoka, M., Tee, W.-L., Witten, C., Zhang, H., and **Zhu, Y.**, **2025**. *JWST COSMOS-3D: Spectroscopic Census and Luminosity Function of [O III] Emitters at 6.75<z<9.05 in COSMOS*, arXiv e-prints, arXiv:2510.11373.

58. Danhaive, A. L., Tacchella, S., McClymont, W., Robertson, B., Carniani, S., Carreira, C., Egami, E., Bunker, A. J., Curtis-Lake, E., Eisenstein, D. J., Ji, Z., Johnson, B. D., Rieke, M., Villanueva, N. C., Willmer, C. N. A., Willott, C., Wu, Z., and **Zhu, Y.**, **2025**. *Beyond the stars: Linking Hα sizes, kinematics, and star formation in galaxies at $z \approx 4-6$ with JWST grism surveys and* `geko`, arXiv e-prints, arXiv:2510.06315.

57. Wu, Z., Eisenstein, D. J., Johnson, B. D., Jakobsen, P., Alberts, S., Arribas, S., Baker, W. M., Bunker, A. J., Carniani, S., Charlot, S., Chevallard, J., Curti, M., Curtis-Lake, E., D'Eugenio, F., Hainline, K., Helton, J. M., Hsiao, T. Y.-Y., Ji, X., Ji, Z., Looser, T. J., Rieke, G., Rinaldi, P., Robertson, B., Scholtz, J., Sun, F., Tacchella, S., Trussler, J. A. A., Williams, C. C., Willmer, C. N. A., Willott, C., Witstok, J., and **Zhu, Y.**, **2025**. *JADES-GS-z14-1: A Compact, Faint Galaxy at z ≈ 14 with Weak Metal Lines from Extremely Deep JWST MIRI, NIRCam, and NIRSpec Observations*, The Astrophysical Journal, 992, 212.

56. Rinaldi, P., Bonaventura, N., Rieke, G. H., Alberts, S., Caputi, K. I., Baker, W. M., Baum, S., Bhatawdekar, R., Bunker, A. J., Carniani, S., Curtis-Lake, E., D'Eugenio, F., Egami, E., Ji, Z., Johnson, B. D., Hainline, K., Helton, J. M., Lin, X., Lyu, J., Ma, Z., Maiolino, R., Pérez-González, P. G., Rieke, M., Robertson, B. E., Shivaei, I., Stone, M., Sun, Y., Tacchella, S., Übler, H., Williams, C. C., Willmer, C. N. A., Willott, C., Zhang, J., and **Zhu, Y.**, **2025**. *Not Just a Dot: The Complex UV Morphology and Underlying Properties of Little Red Dots*, The Astrophysical Journal, 992, 71.

55. Woodrum, C., Shivaei, I., Witstok, J., Saxena, A., Simmonds, C., Scholtz, J., Bhatawdekar, R., Bunker, A. J., Carniani, S., Charlot, S., Curti, M., Curtis-Lake, E., Chevallard, J., D'Eugenio, F., Hainline, K., Helton, J. M., Maiolino, R., Perna, M., Rinaldi, P., Robertson, B., Straughn, A., Sun, Y., Tacchella, S., Williams, C. C., Willott, C., and **Zhu, Y.**, **2025**. *JADES: The Star Formation and Dust Attenuation Properties of Galaxies at 3<z<7*, arXiv e-prints, arXiv:2510.00235.

54. Liu, W., Fan, X., Li, H., Green, R., Yang, J., Jin, X., Lyu, J., Pudoka, M., **Zhu, Y.**, Banados, E., Belladitta, S., Connor, T., Costa, T., Decarli, R., Eilers, A.-C., Jun, H., Marshall, M. A., Mazzucchelli, C., Schindler, J.-T., Shen, Y., Veilleux, S., Wolf, J., Zhang, H., Zhuang, M., Zou, S., and Li, M., **2025**. *Frequent Extreme Galaxy-scale Outflows among Luminous Early Quasars*, arXiv e-prints, arXiv:2509.08793.

53. D'Eugenio, F., Helton, J. M., Hainline, K., Sun, F., Maiolino, R., Pérez-González, P. G., Juodžbalis, I., Arribas, S., Bunker, A. J., Carniani, S., Curtis-Lake, E., Egami, E., Eisenstein, D. J., Johnson, B. D., Robertson, B., Tacchella, S., Willmer, C. N. A., Willott, C., Baker, W. M., Danhaive, A. L., Duan, Q., Fudamoto, Y., Jones, G. C., Lin, X., Liu, W., Perna, M., Puskás, D., Rinaldi, P., Scholtz, J., Sun, Y., Trussler, J. A. A., Übler, H., Venturi, G., Williams, C. C., and **Zhu, Y.**, **2025**. *JADES and SAPPHIRES: galaxy metamorphosis amidst*

a huge, luminous emission-line region, Monthly Notices of the Royal Astronomical Society, 542, 960.

52. Hashemi, S., Becker, G. D., **Zhu, Y.**, and Hong, H., **2025**. *Lyα emission from [O III] emitters near reionization: the role of environment in galaxy Lyα detection*, Monthly Notices of the Royal Astronomical Society, 542, 104.

51. Dong, X., **Zhu, Y.**, Rieke, M., Rieke, G., Li, X., Behroozi, P., Ma, H., Meng, R., Mao, Z., and Sun, Z., **2025**. *A Promise for the JWST era: Massive black holes directly collapsed from wave dark matter haloes, and Star formation in and around their accretion flows*, arXiv e-prints, arXiv:2508.09258.

50. Cohon, J., Cain, C., Windhorst, R., D'Aloisio, A., Carleton, T., and **Zhu, Y.**, **2025**. *A long time ago in an LAE far, far away: a signpost of early reionization or a nascent AGN at $z = 13?$*, arXiv e-prints, arXiv:2508.05739.

49. Ji, Z., Alberts, S., **Zhu, Y.**, Vanzella, E., Giavalisco, M., Hainline, K., Baker, W. M., Bunker, A. J., Helton, J. M., Lyu, J., Rinaldi, P., Robertson, B., Simmonds, C., Tacchella, S., Williams, C. C., Willmer, C. N. A., and Witstok, J., **2025**. *The Importance of Dust Distribution in Ionizing-photon Escape: NIRCam and MIRI Imaging of a Lyman Continuum-emitting Galaxy at $z \sim 3.8$*, The Astrophysical Journal, 988, L69.

48. Rinaldi, P., Rieke, G. H., Wu, Z., Gilbert, C. J. E., Pacucci, F., Barchiesi, L., Alberts, S., Carniani, S., Bunker, A. J., Bhatawdekar, R., D'Eugenio, F., Ji, Z., Johnson, B. D., Hainline, K., Kokorev, V., Kumari, N., Iani, E., Lyu, J., Maiolino, R., Parlanti, E., Robertson, B. E., Sun, Y., Vignali, C., Williams, C. C., Willmer, C. N. A., and **Zhu, Y.**, **2025**. *Beyond the Dot: an LRD-like nucleus at the Heart of an IR-Bright Galaxy and its implications for high-redshift LRDs*, arXiv e-prints, arXiv:2507.17738.

47. Laseter, I. H., Maseda, M. V., Simmonds, C., Endsley, R., Stark, D., Bunker, A. J., Bhatawdekar, R., Boyett, K., Cameron, A. J., Carniani, S., Curti, M., Ji, Z., Rinaldi, P., Saxena, A., Tacchella, S., Willott, C., Witstok, J., and **Zhu, Y.**, **2025**. *Efficient Ionizers with Low Hβ + [O III] Equivalent Widths: JADES Spectroscopy of a Peculiar High-redshift Population*, The Astrophysical Journal, 988, 73.

46. Rieke, G. H., Buiten, V. A., Goldberg, C. E., Morrison, J., van der Werf, P., Alonso-Herrero, A., Alberts, S., Bonaventura, N., Ji, Z., Lyu, J., Rinaldi, P., Stone, M. A., Sun, Y., and **Zhu, Y.**, **2025**. *Low Accretion Rates in Black Holes in Late-stage Merger Ultraluminous Infrared Galaxies*, The Astrophysical Journal, 988, 17.

45. Fu, S., Sun, F., Jiang, L., Lin, X., Diego, J. M., Furtak, L. J., Jauzac, M., Koekemoer, A. M., Li, M., Oguri, M., Patel, N. R., Willmer, C. N. A., Windhorst, R. A., Zitrin, A., Bauer, F. E., Chen, C.-C., Chen, W., Cheng, C., Conselice, C. J., Eisenstein, D. J., Egami, E., Espada, D., Fan, X., Fujimoto, S., Hsiao, T. Y.-Y., Jin, X., Kohno, K., Lagattuta, D. J., Li, Z., Liu, W., Miralda-Escudé, J., Ning, Y., Tacchella, S., Tee, W. L., Umehata, H., Wang, F., Yan, H., and **Zhu, Y.**, **2025**. *Medium-band Astrophysics with the Grism of NIRCam In Frontier Fields (MAGNIF): Spectroscopic Census of Hα Luminosity Functions and Cosmic Star Formation at $z \sim 4.5$ and 6.3*, The Astrophysical Journal, 987, 186.

44. Marcelin, L. C., Champagne, J. B., Wang, F., Fan, X., Pudoka, M., Tee, W. L., and **Zhu, Y.**, **2025**. *Enhanced Merger Fractions in a Reionization-era Protocluster*, Research Notes of the American Astronomical Society, 9, 133.

43. Hsiao, T. Y.-Y., Sun, F., Lin, X., Coe, D., Egami, E., Eisenstein, D. J., Fudamoto, Y., Bunker, A. J., Fan, X., Harikane, Y., Helton, J. M., Kakiichi, K., Liu, Y., Liu, W., Maiolino, R., Ouchi, M., Tee, W. L., Wang, F., Wu, Y., Xu, Y., Yang, J., and **Zhu, Y.**, **2025**. *SAPPHIRES: Extremely Metal-Poor Galaxy Candidates with* $12 + \log(\mathrm{O/H}) < 7.0$ *at* $z \sim 5 - 7$ *from Deep JWST/NIRCam Grism Observations*, arXiv e-prints, arXiv:2505.03873.

42. Helton, J. M., Rieke, G. H., Alberts, S., Wu, Z., Eisenstein, D. J., Hainline, K. N., Carniani, S., Ji, Z., Baker, W. M., Bhatawdekar, R., Bunker, A. J., Cargile, P. A., Charlot, S., Chevallard, J., D'Eugenio, F., Egami, E., Johnson, B. D., Jones, G. C., Lyu, J., Maiolino, R., Pérez-González, P. G., Rieke, M. J., Robertson, B., Saxena, A., Scholtz, J., Shivaei, I., Sun, F., Tacchella, S., Whitler, L., Williams, C. C., Willmer, C. N. A., Willott, C., Witstok, J., and **Zhu, Y.**, **2025**. *Photometric detection at 7.7* μ*m of a galaxy beyond redshift 14 with JWST/MIRI*, Nature Astronomy, 9, 729.

41. Baker, W. M., Lim, S., D'Eugenio, F., Maiolino, R., Ji, Z., Arribas, S., Bunker, A. J., Carniani, S., Charlot, S., de Graaff, A., Hainline, K., Looser, T. J., Lyu, J., Rinaldi, P., Robertson, B., Schaller, M., Schaye, J., Scholtz, J., Übler, H., Williams, C. C., Willmer, C. N. A., Willott, C., and **Zhu, Y.**, **2025**. *The abundance and nature of high-redshift quiescent galaxies from JADES spectroscopy and the FLAMINGO simulations*, Monthly Notices of the Royal Astronomical Society, 539, 557.

40. Baker, W. M., D'Eugenio, F., Maiolino, R., Bunker, A. J., Simmonds, C., Tacchella, S., Witstok, J., Arribas, S., Carniani, S., Charlot, S., Chevallard, J., Curti, M., Curtis-Lake, E., Jones, G. C., Kumari, N., Rinaldi, P., Robertson, B., Williams, C. C., Willott, C., and **Zhu, Y.**, **2025**. *Zapped then napped? A rapidly quenched remnant leaker candidate with a steep spectroscopic* β*_{UV} slope at* z *= 8.5*, Astronomy and Astrophysics, 697, A90.

39. Qin, Y., Mesinger, A., Prelogović, D., Becker, G., Bischetti, M., Bosman, S., Davies, F., D'Odorico, V., Gaikwad, P., Haehnelt, M., Keating, L., Lai, S., Ryan-Weber, E., Satyavolu, S., Walter, F., and **Zhu, Y.**, **2025**. *Percent-level timing of reionisation: Self-consistent, implicit-likelihood inference from XQR-30+ Ly*α *forest data*, Publications of the Astronomical Society of Australia, 42, e049.

38. Sun, Y., Rieke, G. H., Lyu, J., Stone, M. A., Ji, Z., Rinaldi, P., Willmer, C. N. A., and **Zhu, Y.**, **2025**. *Evolution of the M_*—M_{BH} Relation from* $z \sim 6$ *to the Present Epoch*, The Astrophysical Journal, 983, 165.

37. Fudamoto, Y., Helton, J. M., Lin, X., Sun, F., Behroozi, P., Hsiao, T. Y.-Y., Egami, E., Bunker, A. J., Harikane, Y., Ouchi, M., Liu, Y., Liu, W., Maiolino, R., Ji, Z., Jin, X., Tee, W. L., Wang, F., Willmer, C. N. A., Xu, Y., and **Zhu, Y.**, **2025**. *SAPPHIRES: A Galaxy Over-Density in the Heart of Cosmic Reionization at* $z = 8.47$, arXiv e-prints, arXiv:2503.15597.

36. Sun, F., Fudamoto, Y., Lin, X., Helton, J. M., Hsiao, T. Y.-Y., Egami, E., Akhtarkavan, A., Bunker, A. J., Cai, Z., DeCoursey, C., Eisenstein, D. J., Fan, X., Harikane, Y., Ji, Z., Jin,

X., Liu, W., Liu, Y., Ma, Z., Maiolino, R., Ouchi, M., Tee, W. L., Wang, F., Willmer, C. N. A., Wu, Y., Xu, Y., Yang, J., Zhang, J., and **Zhu, Y.**, **2025**. *Slitless Areal Pure-Parallel HIgh-Redshift Emission Survey (SAPPHIRES): Early Data Release of Deep JWST/NIRCam Images and Spectra in MACS J0416 Parallel Field*, arXiv e-prints, arXiv:2503.15587.

35. Witstok, J., Jakobsen, P., Maiolino, R., Helton, J. M., Johnson, B. D., Robertson, B. E., Tacchella, S., Cameron, A. J., Smit, R., Bunker, A. J., Saxena, A., Sun, F., Alberts, S., Arribas, S., Baker, W. M., Bhatawdekar, R., Boyett, K., Cargile, P. A., Carniani, S., Charlot, S., Chevallard, J., Curti, M., Curtis-Lake, E., D'Eugenio, F., Eisenstein, D. J., Hainline, K. N., Jones, G. C., Kumari, N., Maseda, M. V., Pérez-González, P. G., Rinaldi, P., Scholtz, J., Übler, H., Williams, C. C., Willmer, C. N. A., Willott, C., and **Zhu, Y.**, **2025**. *Witnessing the onset of reionization through Lyman-α emission at redshift 13*, Nature, 639, 897.

34. Fudamoto, Y., Sun, F., Diego, J. M., Dai, L., Oguri, M., Zitrin, A., Zackrisson, E., Jauzac, M., Lagattuta, D. J., Egami, E., Iani, E., Windhorst, R. A., Abe, K. T., Bauer, F. E., Bian, F., Bhatawdekar, R., Broadhurst, T. J., Cai, Z., Chen, C.-C., Chen, W., Cohen, S. H., Conselice, C. J., Espada, D., Foo, N., Frye, B. L., Fujimoto, S., Furtak, L. J., Golubchik, M., Hsiao, T. Y.-Y., Jolly, J.-B., Kawai, H., Kelly, P. L., Koekemoer, A. M., Kohno, K., Kokorev, V., Li, M., Li, Z., Lin, X., Magdis, G. E., Meena, A. K., Niemiec, A., Nabizadeh, A., Richard, J., Steinhardt, C. L., Wu, Y., **Zhu, Y.**, and Zou, S., **2025**. *Identification of more than 40 gravitationally magnified stars in a galaxy at redshift 0.725*, Nature Astronomy, 9, 428.

33. D'Eugenio, F., Cameron, A. J., Scholtz, J., Carniani, S., Willott, C. J., Curtis-Lake, E., Bunker, A. J., Parlanti, E., Maiolino, R., Willmer, C. N. A., Jakobsen, P., Robertson, B. E., Johnson, B. D., Tacchella, S., Cargile, P. A., Rawle, T., Arribas, S., Chevallard, J., Curti, M., Egami, E., Eisenstein, D. J., Kumari, N., Looser, T. J., Rieke, M. J., Rodríguez Del Pino, B., Saxena, A., Übler, H., Venturi, G., Witstok, J., Baker, W. M., Bhatawdekar, R., Bonaventura, N., Boyett, K., Charlot, S., Danhaive, A. L., Hainline, K. N., Hausen, R., Helton, J. M., Ji, X., Ji, Z., Jones, G. C., Juodžbalis, I., Maseda, M. V., Pérez-González, P. G., Perna, M., Puskás, D., Shivaei, I., Silcock, M. S., Simmonds, C., Smit, R., Sun, F., Villanueva, N. C., Williams, C. C., and **Zhu, Y.**, **2025**. *JADES Data Release 3: NIRSpec/Microshutter Assembly Spectroscopy for 4000 Galaxies in the GOODS Fields*, The Astrophysical Journal Supplement Series, 277, 4.

32. Kaur, B., Kanekar, N., Neeleman, M., **Zhu, Y.**, Prochaska, J. X., Rafelski, M., and Becker, G., **2025**. *A Massive H I-absorption-selected Galaxy at $z \asymp 2.356$*, The Astrophysical Journal, 982, L26.

31. Lyu, J., Rieke, G. H., Stone, M., Morrison, J., Alberts, S., Jin, X., **Zhu, Y.**, Liu, W., and Yang, J., **2025**. *Fading Light, Fierce Winds: JWST Snapshot of a Sub-Eddington Quasar at Cosmic Dawn*, The Astrophysical Journal, 981, L20.

30. Ma, H.-X., Takeuchi, T. T., Cooray, S., and **Zhu, Y.**, **2025**. *sOPTICS: a modified density-based algorithm for identifying galaxy groups/clusters and brightest cluster galaxies*, Monthly Notices of the Royal Astronomical Society, 537, 1504.

29. Hainline, K. N., Maiolino, R., Juodžbalis, I., Scholtz, J., Übler, H., D'Eugenio, F., Helton, J. M., Sun, Y., Sun, F., Robertson, B., Tacchella, S., Bunker, A. J., Carniani, S., Charlot, S.,

Curtis-Lake, E., Egami, E., Johnson, B. D., Lin, X., Lyu, J., Pérez-González, P. G., Rinaldi, P., Silcock, M. S., Venturi, G., Williams, C. C., Willmer, C. N. A., Willott, C., Zhang, J., and **Zhu, Y.**, **2025**. *An Investigation into the Selection and Colors of Little Red Dots and Active Galactic Nuclei*, The Astrophysical Journal, 979, 138.

28. Jones, G. C., Bunker, A. J., Saxena, A., Arribas, S., Bhatawdekar, R., Boyett, K., Cameron, A. J., Carniani, S., Charlot, S., Curtis-Lake, E., Hainline, K., Johnson, B. D., Kumari, N., Maseda, M. V., Rix, H.-W., Robertson, B. E., Tacchella, S., Übler, H., Williams, C. C., Willott, C., Witstok, J., and **Zhu, Y.**, **2025**. *JADES: measuring reionization properties using Lyman-alpha emission*, Monthly Notices of the Royal Astronomical Society, 536, 2355.

27. Sun, Y., Lyu, J., Rieke, G. H., Ji, Z., Sun, F., **Zhu, Y.**, Bunker, A. J., Cargile, P. A., Circosta, C., D'Eugenio, F., Egami, E., Hainline, K., Helton, J. M., Rinaldi, P., Robertson, B. E., Scholtz, J., Shivaei, I., Stone, M. A., Tacchella, S., Williams, C. C., Willmer, C. N. A., and Willott, C., **2025**. *No Evidence for a Significant Evolution of M_•—M. Relation in Massive Galaxies up to $z \sim 4$*, The Astrophysical Journal, 978, 98.

26. Simmonds, C., Tacchella, S., Hainline, K., Johnson, B. D., Puskás, D., Robertson, B., Baker, W. M., Bhatawdekar, R., Boyett, K., Bunker, A. J., Cargile, P. A., Carniani, S., Chevallard, J., Curti, M., Curtis-Lake, E., Ji, Z., Jones, G. C., Kumari, N., Laseter, I., Maiolino, R., Maseda, M. V., Rinaldi, P., Stoffers, A., Übler, H., Villanueva, N. C., Williams, C. C., Willott, C., Witstok, J., and **Zhu, Y.**, **2024**. *Ionizing properties of galaxies in JADES for a stellar mass complete sample: resolving the cosmic ionizing photon budget crisis at the Epoch of Reionization*, Monthly Notices of the Royal Astronomical Society, 535, 2998.

25. Alberts, S., Lyu, J., Shivaei, I., Rieke, G. H., Pérez-González, P. G., Bonaventura, N., **Zhu, Y.**, Helton, J. M., Ji, Z., Morrison, J., Robertson, B. E., Stone, M. A., Sun, Y., Williams, C. C., and Willmer, C. N. A., **2024**. *SMILES Initial Data Release: Unveiling the Obscured Universe with MIRI Multiband Imaging*, The Astrophysical Journal, 976, 224.

24. Saxena, A., Cameron, A. J., Katz, H., Bunker, A. J., Chevallard, J., D'Eugenio, F., Arribas, S., Bhatawdekar, R., Boyett, K., Cargile, P. A., Carniani, S., Charlot, S., Curti, M., Curtis-Lake, E., Hainline, K., Ji, Z., Johnson, B. D., Jones, G. C., Kumari, N., Laseter, I., Maseda, M. V., Robertson, B., Simmonds, C., Tacchella, S., Ubler, H., Williams, C. C., Willott, C., Witstok, J., and **Zhu, Y.**, **2024**. *Hitting the slopes: A spectroscopic view of UV continuum slopes of galaxies reveals a reddening at $z > 9.5$*, arXiv e-prints, arXiv:2411.14532.

23. Jiang, D., Onoue, M., Jiang, L., Lai, S., Bañados, E., Becker, G. D., Bischetti, M., Bosman, S. E. I., Davies, R. L., D'Odorico, V., Farina, E. P., Haehnelt, M. G., Mazzucchelli, C., Schindler, J.-T., Walter, F., and **Zhu, Y.**, **2024**. *No Redshift Evolution in the Fe II/Mg II Flux Ratios of Quasars across Cosmic Time*, The Astrophysical Journal, 975, 214.

22. Shivaei, I., Alberts, S., Florian, M., Rieke, G., Wuyts, S., Bodansky, S., Bunker, A. J., Cameron, A. J., Curti, M., D'Eugenio, F., Dudzevičiūtė, U., Ji, Z., Johnson, B. D., Kramarenko, I., Lyu, J., Matthee, J., Morrison, J., Naidu, R., Pérez-González, P. G., Reddy, N., Robertson, B., Sun, Y., Tacchella, S., Whitaker, K., Williams, C. C., Willmer, C. N. A., Witstok, J., Xiao, M., and **Zhu, Y.**, **2024**. *A new census of dust and polycyclic aromatic hydrocarbons at $z = 0.7$—2 with JWST MIRI*, Astronomy and Astrophysics, 690, A89.

21. Ji, Z., Williams, C. C., Rieke, G. H., Lyu, J., Alberts, S., Sun, F., Helton, J. M., Rieke, M., Shivaei, I., D'Eugenio, F., Tacchella, S., Robertson, B., **Zhu, Y.**, Maiolino, R., Bunker, A. J., Sun, Y., and Willmer, C. N. A., **2024**. *Extended hot dust emission around the earliest massive quiescent galaxy*, arXiv e-prints, arXiv:2409.17233.

20. Becker, G. D., Bolton, J. S., **Zhu, Y.**, and Hashemi, S., **2024**. *Damping wing absorption associated with a giant Ly α trough at $z < 6$: direct evidence for late-ending reionization*, Monthly Notices of the Royal Astronomical Society, 533, 1525.

19. Stone, M. A., Alberts, S., Rieke, G. H., Bunker, A. J., Lyu, J., Pérez-González, P. G., Shivaei, I., and **Zhu, Y.**, **2024**. *5—25 μm Galaxy Number Counts from Deep JWST Data*, The Astrophysical Journal, 972, 62.

18. Spina, B., Bosman, S. E. I., Davies, F. B., Gaikwad, P., and **Zhu, Y.**, **2024**. *Damping wings in the Lyman-α forest: A model-independent measurement of the neutral fraction at 5.4 < z < 6.1*, Astronomy and Astrophysics, 688, L26.

17. Wolfson, M., Hennawi, J. F., Bosman, S. E. I., Davies, F. B., Lukić, Z., Becker, G. D., Chen, H., Cupani, G., D'Odorico, V., Eilers, A.-C., Haehnelt, M. G., Keating, L. C., Kulkarni, G., Lai, S., Mesinger, A., Walter, F., and **Zhu, Y.**, **2024**. *Measurements of the $z > 5$ Lyman-α forest flux autocorrelation functions from the extended XQR-30 data set*, Monthly Notices of the Royal Astronomical Society, 531, 3069.

16. Bischetti, M., Choi, H., Fiore, F., Feruglio, C., Carniani, S., D'Odorico, V., Bañados, E., Chen, H., Decarli, R., Gallerani, S., Hlavacek-Larrondo, J., Lai, S., Leighly, K. M., Mazzucchelli, C., Perreault-Levasseur, L., Tripodi, R., Walter, F., Wang, F., Yang, J., Zanchettin, M. V., and **Zhu, Y.**, **2024**. *Multiphase Black Hole Feedback and a Bright [C II] Halo in a LoBAL Quasar at $z \sim 6.6$*, The Astrophysical Journal, 970, 9.

15. Roth, J. T., D'Aloisio, A., Cain, C., Wilson, B., **Zhu, Y.**, and Becker, G. D., **2024**. *The effect of reionization on direct measurements of the mean free path*, Monthly Notices of the Royal Astronomical Society, 530, 5209.

14. Greig, B., Mesinger, A., Bañados, E., Becker, G. D., Bosman, S. E. I., Chen, H., Davies, F. B., D'Odorico, V., Eilers, A.-C., Gallerani, S., Haehnelt, M. G., Keating, L., Lai, S., Qin, Y., Ryan-Weber, E., Satyavolu, S., Wang, F., Yang, J., and **Zhu, Y.**, **2024**. *IGM damping wing constraints on the tail end of reionization from the enlarged XQR-30 sample*, Monthly Notices of the Royal Astronomical Society, 530, 3208.

13. Davies, F. B., Bosman, S. E. I., Gaikwad, P., Nasir, F., Hennawi, J. F., Becker, G. D., Haehnelt, M. G., D'Odorico, V., Bischetti, M., Eilers, A.-C., Keating, L. C., Kulkarni, G., Lai, S., Mazzucchelli, C., Qin, Y., Satyavolu, S., Wang, F., Yang, J., and **Zhu, Y.**, **2024**. *Constraints on the Evolution of the Ionizing Background and Ionizing Photon Mean Free Path at the End of Reionization*, The Astrophysical Journal, 965, 134.

12. Gaikwad, P., Haehnelt, M. G., Davies, F. B., Bosman, S. E. I., Molaro, M., Kulkarni, G., D'Odorico, V., Becker, G. D., Davies, R. L., Nasir, F., Bolton, J. S., Keating, L. C., Iršič, V., Puchwein, E., **Zhu, Y.**, Asthana, S., Yang, J., Lai, S., and Eilers, A.-C., **2023**. *Measuring the photoionization rate, neutral fraction, and mean free path of H I ionizing photons at 4.9*

$\leq z \leq$ *6.0 from a large sample of XShooter and ESI spectra*, Monthly Notices of the Royal Astronomical Society, 525, 4093.

11. Christenson, H. M., Becker, G. D., D'Aloisio, A., Davies, F. B., **Zhu, Y.**, Boera, E., Nasir, F., Furlanetto, S. R., and Malkan, M. A., **2023**. *The Relationship between IGM Lyα Opacity and Galaxy Density near the End of Reionization*, The Astrophysical Journal, 955, 138.

10. Mazzucchelli, C., Bischetti, M., D'Odorico, V., Feruglio, C., Schindler, J.-T., Onoue, M., Bañados, E., Becker, G. D., Bian, F., Carniani, S., Decarli, R., Eilers, A.-C., Farina, E. P., Gallerani, S., Lai, S., Meyer, R. A., Rojas-Ruiz, S., Satyavolu, S., Venemans, B. P., Wang, F., Yang, J., and **Zhu, Y.**, **2023**. *XQR-30: Black hole masses and accretion rates of 42 $z \gtrsim 6$ quasars*, Astronomy and Astrophysics, 676, A71.

9. D'Odorico, V., Bañados, E., Becker, G. D., Bischetti, M., Bosman, S. E. I., Cupani, G., Davies, R., Farina, E. P., Ferrara, A., Feruglio, C., Mazzucchelli, C., Ryan-Weber, E., Schindler, J.-T., Sodini, A., Venemans, B. P., Walter, F., Chen, H., Lai, S., **Zhu, Y.**, Bian, F., Campo, S., Carniani, S., Cristiani, S., Davies, F., Decarli, R., Drake, A., Eilers, A.-C., Fan, X., Gaikwad, P., Gallerani, S., Greig, B., Haehnelt, M. G., Hennawi, J., Keating, L., Kulkarni, G., Mesinger, A., Meyer, R. A., Neeleman, M., Onoue, M., Pallottini, A., Qin, Y., Rojas-Ruiz, S., Satyavolu, S., Sebastian, A., Tripodi, R., Wang, F., Wolfson, M., Yang, J., and Zanchettin, M. V., **2023**. *XQR-30: The ultimate XSHOOTER quasar sample at the reionization epoch*, Monthly Notices of the Royal Astronomical Society, 523, 1399.

8. Bischetti, M., Fiore, F., Feruglio, C., D'Odorico, V., Arav, N., Costa, T., Zubovas, K., Becker, G., Bosman, S. E. I., Cupani, G., Davies, R., Eilers, A.-C., Farina, E. P., Ferrara, A., Gaspari, M., Mazzucchelli, C., Onoue, M., Piconcelli, E., Zanchettin, M. V., and **Zhu, Y.**, **2023**. *The Fraction and Kinematics of Broad Absorption Line Quasars across Cosmic Time*, The Astrophysical Journal, 952, 44.

7. Davies, R. L., Ryan-Weber, E., D'Odorico, V., Bosman, S. E. I., Meyer, R. A., Becker, G. D., Cupani, G., Keating, L. C., Bischetti, M., Davies, F. B., Eilers, A.-C., Farina, E. P., Haehnelt, M. G., Pallottini, A., and **Zhu, Y.**, **2023**. *Examining the decline in the C IV content of the Universe over 4.3 $\lesssim z \lesssim$ 6.3 using the E-XQR-30 sample*, Monthly Notices of the Royal Astronomical Society, 521, 314.

6. Davies, R. L., Ryan-Weber, E., D'Odorico, V., Bosman, S. E. I., Meyer, R. A., Becker, G. D., Cupani, G., Bischetti, M., Sebastian, A. M., Eilers, A.-C., Farina, E. P., Wang, F., Yang, J., and **Zhu, Y.**, **2023**. *The XQR-30 metal absorber catalogue: 778 absorption systems spanning 2 $\lesssim z \lesssim$ 6.5*, Monthly Notices of the Royal Astronomical Society, 521, 289.

5. Bosman, S. E. I., Davies, F. B., Becker, G. D., Keating, L. C., Davies, R. L., **Zhu, Y.**, Eilers, A.-C., D'Odorico, V., Bian, F., Bischetti, M., Cristiani, S. V., Fan, X., Farina, E. P., Haehnelt, M. G., Hennawi, J. F., Kulkarni, G., Mesinger, A., Meyer, R. A., Onoue, M., Pallottini, A., Qin, Y., Ryan-Weber, E., Schindler, J.-T., Walter, F., Wang, F., and Yang, J., **2022**. *Hydrogen reionization ends by z = 5.3: Lyman-α optical depth measured by the XQR-30 sample*, Monthly Notices of the Royal Astronomical Society, 514, 55.

4. Lai, S., Bian, F., Onken, C. A., Wolf, C., Mazzucchelli, C., Bañados, E., Bischetti, M., Bosman, S. E. I., Becker, G., Cupani, G., D'Odorico, V., Eilers, A.-C., Fan, X., Farina, E.

P., Onoue, M., Schindler, J.-T., Walter, F., Wang, F., Yang, J., and **Zhu, Y.**, **2022**. *Chemical abundance of $z \sim 6$ quasar broad-line regions in the XQR-30 sample*, Monthly Notices of the Royal Astronomical Society, 513, 1801.

3. Bischetti, M., Feruglio, C., D'Odorico, V., Arav, N., Bañados, E., Becker, G., Bosman, S. E. I., Carniani, S., Cristiani, S., Cupani, G., Davies, R., Eilers, A. C., Farina, E. P., Ferrara, A., Maiolino, R., Mazzucchelli, C., Mesinger, A., Meyer, R. A., Onoue, M., Piconcelli, E., Ryan-Weber, E., Schindler, J.-T., Wang, F., Yang, J., **Zhu, Y.**, and Fiore, F., **2022**. *Suppression of black-hole growth by strong outflows at redshifts 5.8-6.6*, Nature, 605, 244.

2. Becker, G. D., D'Aloisio, A., Christenson, H. M., **Zhu, Y.**, Worseck, G., and Bolton, J. S., **2021**. *The mean free path of ionizing photons at $5 < z < 6$: evidence for rapid evolution near reionization*, Monthly Notices of the Royal Astronomical Society, 508, 1853.

1. Christenson, H. M., Becker, G. D., Furlanetto, S. R., Davies, F. B., Malkan, M. A., **Zhu, Y.**, Boera, E., and Trapp, A., **2021**. *Constraints on the End of Reionization from the Density Fields Surrounding Two Highly Opaque Quasar Sightlines*, The Astrophysical Journal, 923, 87.